

Shaer: Controlled Arabic Poetry Generation with Meter Subform and Semantic Conditioning

Ahmad Abbas

Tamara Fakih

Nour Fakih

Ammar Mohanna

Department of Electrical and Computer Engineering

American University of Beirut, Beirut, Lebanon

{asa108,tmf14,nhf16}@mail.aub.edu

am288@aub.edu.lb

Abstract

Classical Arabic poetry generation requires simultaneously satisfying semantic, linguistic, and fine-grained prosodic constraints. Existing systems typically control broad poetic attributes but do not jointly model semantic intent, meter subform, and poem length. We present SHAER, a controllable Classical Arabic poetry generation framework jointly conditioned on natural-language descriptions, meter subforms, and target hemistich counts. To support this task, we construct an enriched corpus of 116,032 classical Arabic poems derived from Ashaar, containing normalized meter-subform labels and automatically generated, validated semantic descriptions. We then adapt Yehia-7B using QLoRA-based supervised fine-tuning with a completion-only objective. Our evaluation combines automatic assessment of base-meter conformity, requested-subform adherence, and length control with three LLM judges, blinded human evaluation, and memorization analysis. SHAER achieves 95.17% base-meter accuracy, 91.75% poem-level meter-subform accuracy, and 83.40% exact count accuracy. Relative to its untuned foundation model, these results represent gains of 68.68, 57.77, and 38.93 percentage points, respectively; SHAER also attains the highest base-meter accuracy among all evaluated systems. Multi-LLM evaluation and a blinded human assessment of top-ranked outputs further indicate competitive semantic and literary quality. Finally, analysis of all 3,481 test generations finds no exact copies from the training corpus or paired source poems. Code, models, and datasets are publicly available.¹

1 Introduction

Classical Arabic poetry is a central component of Arabic literary heritage, making controlled generation both a technically challenging problem and a culturally grounded Arabic NLP task. Unlike

conventional language generation tasks, successful poetry generation must simultaneously satisfy semantic, stylistic, and prosodic constraints. Beyond producing fluent and coherent text, a generated poem must adhere to a prescribed meter while faithfully expressing the intended meaning. These requirements make Arabic poetry generation a particularly demanding controllable generation problem. A brief overview of Classical Arabic poetic structure, meter, and meter subforms is provided in Appendix A.

Despite recent progress in Arabic language generation and computational poetry, fine-grained control over generated Arabic poetry remains challenging. Existing generation settings largely operate over broad structural or categorical attributes, leaving important dimensions of poetic control insufficiently addressed. In particular, jointly controlling natural-language semantic intent, fine-grained meter subforms, and explicit poem length remains underexplored. This limits the ability to generate poems that simultaneously realize a desired meaning and satisfy precise prosodic and structural requirements.

To address these gaps, we introduce SHAER, a controllable Arabic poetry generation framework that jointly models semantic intent and poetic structure. By conditioning generation on natural-language descriptions, meter subforms, and target hemistich counts, SHAER enables fine-grained control over meaning, meter, and poem structure. Figure 1 presents representative poems generated by SHAER from prompts drawn from the test split of the released dataset.

Our contributions are as follows: (1) We introduce a controllable Arabic poetry generation task conditioned jointly on semantic descriptions, meter subforms, and requested poem length, enabling fine-grained control over semantic and structural properties. (2) We construct an enriched Arabic poetry corpus derived from Ashaar, comprising

¹Code: <https://github.com/AhmaddAbbass/ShaeR>. Models and datasets: <https://huggingface.co/ShaeR-AI>.

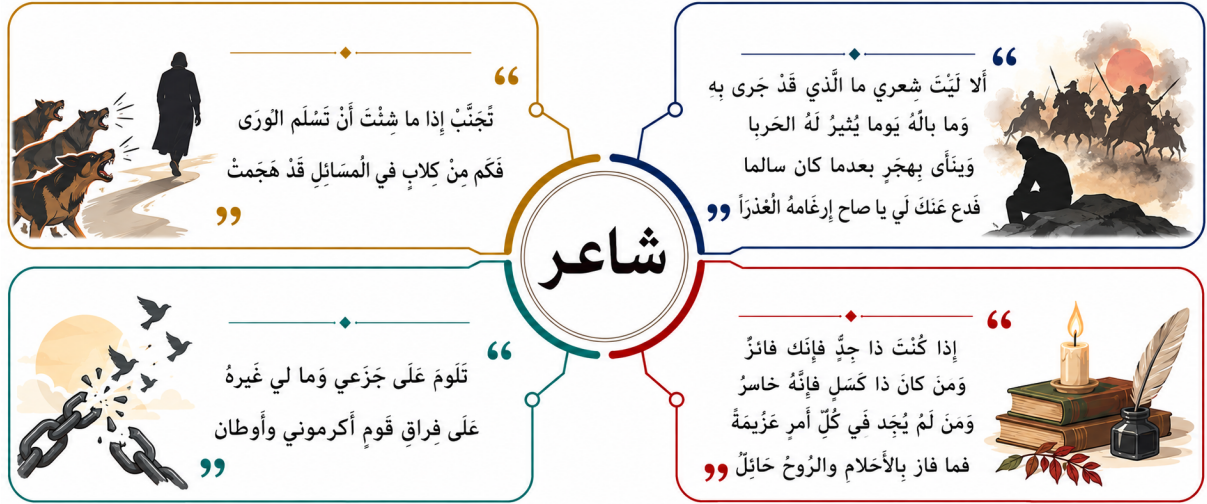


Figure 1: Example poems generated by Shaer. Each poem is generated from a semantic description, meter specification, and length constraint derived from a held-out test example.

116,032 classical poems annotated with normalized meter subforms and semantic descriptions, together with the complete data processing pipeline. (3) We develop SHAER, a controllable Arabic poetry generation model built by adapting Yehia-7B, a strong Arabic-focused language model, through parameter-efficient fine-tuning and explicit conditioning on semantic and prosodic constraints. (4) We introduce a multi-perspective evaluation framework combining automatic assessment of base-meter conformity, meter-subform adherence, and length control with three LLM judges, blinded human evaluation, and memorization and copying analysis.

Experimental results show that SHAER attains the highest base-meter accuracy among all evaluated systems and achieves high requested-subform adherence. Relative to its untuned foundation model, Yehia-7B, SHAER provides markedly stronger base-meter, meter-subform, and length control. Evaluation with three LLM judges and a blinded human assessment of top-ranked outputs further indicates that these gains in controllability are achieved while retaining competitive semantic and literary quality.

2 Related Work

2.1 Arabic Poetry Generation

Early neural approaches to Arabic poetry generation relied on task-specific architectures. Talafha and Rekabdar (2019) combined keyword-based semantic guidance with Bi-GRU generation and hierarchical attention over previously generated verses, while their later work incorporated phonetic CNN

subword embeddings and keyword expansion to improve metrical conformity and semantic consistency (Talafha and Rekabdar, 2021).

Pretrained language models subsequently enabled transfer-based approaches. Beheitt and Ben Haj Hmida (2022) pretrained GPT-2 on Arabic text and fine-tuned it for poetry generation, while Ab-boushi and Azzeh (2023) fine-tuned AraGPT2 on a large poetry corpus with explicit meter and rhyme conditioning.

Recent work has expanded both resources and controllability. Ashaar (Alyafeai et al., 2023) introduced datasets and pretrained models for Arabic poetry analysis and conditional generation. Fanar-Diwan (FANAR Team et al., 2026) supports controllable classical Arabic poetry generation conditioned on meter, topic, era, poet, and rhyme. From an instruction-following perspective, Sadallah et al. (2026) introduced a large-scale resource spanning Modern Standard Arabic and multiple dialects and supporting poetry generation, continuation, revision, and analysis under user-specified constraints.

Building on this progression, SHAER offers a complementary approach to controllable classical Arabic poetry generation through joint conditioning on natural-language semantic descriptions, meter subforms, and explicit poem length.

2.2 Arabic Poetry Analysis and Evaluation

Arabic poetry analysis has extensively addressed meter identification. Al-Shaibani et al. (2020) proposed a character-based bidirectional recurrent model for classifying poetry into 14 meters without

traditional Arudi representation, while [Abandah et al. \(2022\)](#) developed recurrent models covering the 16 classical meters and prose, alongside automatic diacritization. We reproduce the latter meter-classification architecture to assess base-meter conformity.

For finer-grained prosody, [Qarah \(2024\)](#) introduced AraPoemBERT, a poetry-specific language model evaluated on several analysis tasks, including meter-subform classification. We reproduce this setting to assess subform adherence in controllable generation.

At the semantic level, [Al Ghallabi et al. \(2025\)](#) introduced Fann or Flop, a benchmark for LLM understanding of Arabic poetry across semantic, figurative, prosodic, and cultural dimensions. Our evaluation similarly adopts a multidimensional perspective, combining automatic assessment of meter, subform, and length control with LLM-based and human judgments of semantic and literary quality.

3 Dataset

3.1 Source Corpus

Our work builds on Ashaar ([Alyafeai et al., 2023](#)), a large-scale Arabic poetry resource containing 254,630 poems from multiple online repositories. Ashaar supports several poetry-related tasks and provides poem-level metadata including meter, theme, poet identity, era, and descriptions. However, its metadata organization and description coverage are not directly suited to controllable generation based on fine-grained poetic structure and semantic intent. Appendix B provides a detailed analysis of the original corpus.

3.2 Corpus Construction

We construct an enriched corpus for controllable Arabic poetry generation through multi-stage filtering and normalization. We retain only classical Arabic poetry in the canonical Khalilian meters, removing colloquial or non-classical forms, malformed poems, and entries with invalid verse structure. To align with the 1,024-token maximum sequence length used for supervised fine-tuning, we exclude poems exceeding twenty verses. A post-tokenization audit using the final instruction format found that only 4 of 109,070 training sequences (0.0037%) exceeded this limit, by 5–25 tokens, with none in the validation or test sets. Finally, we remove insufficiently represented meter categories. Appendix C details the cleaning, normalization,

and filtering procedures.

3.3 Meter Subform Annotation and Validation

All 116,032 retained poems inherit their base-meter labels from Ashaar. Explicit meter-subform metadata was available for 5,106 poems (4.4%); we normalized and retained these labels without independent re-annotation. The remaining 110,926 poems (95.6%) were assigned the default **تام** (Complete) label, corresponding to the canonical realization of their base meter in classical Arabic prosody.

We assessed this default assignment in two ways. First, an author formally trained in Arabic prosody manually inspected 1,000 default-labeled poems, confirming the Complete form in 97.2% of cases. Second, we reproduced the AraPoemBERT meter-subform classification setting ([Qarah, 2024](#)), achieving 97.38% accuracy compared with the originally reported 97.79%. Applied to all 110,926 default-labeled poems, the classifier predicted Complete for 103,748 (93.53%). Agreement exceeded 98.7% for nine of thirteen base meters, with disagreements concentrated in four meters affected by classifier limitations in class coverage or reported per-class performance. We use these predictions only to assess the default-assignment convention, not to replace corpus labels; complete per-meter results and discussion are provided in Appendix D.1.

After normalization and default assignment, all poems have meter-subform annotations, yielding 21 distinct meter-subform combinations whose distribution is reported in Appendix D.2.

3.4 Semantic Description Generation

Ashaar descriptions are missing or insufficiently informative for much of the corpus, limiting their use as semantic conditioning signals (Appendix B). We therefore generate a semantic description for every poem.

We compare three generation configurations differing in model and prompting strategy. For each, three native Arabic-speaking annotators independently evaluate a stratified sample of 600 descriptions using a predefined Pass/Fail rubric. The best-performing configuration, based on Qwen3.5-35B-A3B ([Qwen Team, 2026a](#)), achieves a 92.8% pass rate and is selected for corpus-wide generation. Configuration and human-evaluation details are provided in Appendix E.1.

The selected configuration uses deterministic decoding to generate concise descriptions capturing

Meter	Sub-form	Requested Hemistichs	Description	Poem Hemistichs
الوافر	تام	4	القصيدة تتحدث عن ليلٍ طويلٍ كليل يوم القيامة، جعل السهر مستحيلاً، لدرجة أن الشاعر كاد يظن في سهادته أن الله لم يخلق نهراً.	وَلَيْلٌ مِثْلُ يَوْمِ الْحَشْرِ طَوَّلاً مَنْعَتْ بِهِ مِنَ السَّهْرِ الْقَرَارَا وَكَدَتْ أَظُنُّ فِيهِ مِنْ سُهَادِي بِأَنَّ اللَّهَ لَمْ يَخْلُقْ نَهَارَا

Table 1: Example Shaer dataset instance with its conditioning signals: base meter, meter subform, target hemistich count, and semantic description.

each poem’s meaning, imagery, and tone while excluding references to meter, rhyme, or length. The resulting descriptions undergo automatic structural validation and corpus-wide quality checks. Appendices E.2 and E.3 provide full prompting, decoding, and verification details.

3.5 Final Dataset Statistics

The final corpus pairs each poem with its base meter, meter subform, target hemistich count, and generated semantic description, jointly supporting control over poetic form and semantic content. It comprises 116,032 poems spanning 13 classical meters and 21 meter–subform combinations, providing substantially finer-grained metrical supervision than the coarse meter categories commonly used in prior Arabic poetry generation systems. Appendix F summarizes the dataset characteristics, while Table 1 illustrates the enriched metadata and controlled generation format.

3.6 Data Splits

We partition the corpus into train (94%), validation (3%), and test (3%) splits, stratified by a composite key of base meter, meter subform, and length bucket to preserve poetic-structure distributions across partitions. Length buckets are defined by poem length as 1–3, 4–6, 7–10, and 11–20 verses.

4 Methodology

4.1 Task Formulation

We formulate Arabic poetry generation as a controllable generation task. Given a conditioning tuple

$$x = (m, s, n, d),$$

where m is the base meter, s the meter subform, n the requested hemistich count, and d a natural-

language description of the desired content, the model generates

$$y = (y_1, \dots, y_T),$$

where y_t is the t -th generated token and T the total number of generated tokens. The objective is to model $P(y | x)$ such that the generated poem satisfies the specified semantic, prosodic, and length constraints.

4.2 Base Model Selection

Model selection is informed by AraLingBench (Zbib et al., 2026), a human-annotated benchmark of Arabic linguistic competence across grammar, morphology, spelling, syntax, and reading comprehension, capabilities closely aligned with classical Arabic poetry generation.

Yehia-7B (Navid-AI, 2025), a 7B-parameter Arabic-focused decoder-only language model, achieved the strongest overall performance among the evaluated models. We therefore select Navid-AI/Yehia-7B-² as the foundation model for SHAER. Its Apache-2.0 license also facilitates public release of adapters and derived resources.

4.3 Supervised Fine-Tuning

We adapt Yehia-7B through supervised fine-tuning on the enriched corpus training split using a completion-only causal language modeling objective. Given prompt x and target completion y , the loss is computed only over poem tokens:

$$\mathcal{L}(\theta) = - \sum_{t \in \mathcal{C}} \log p_{\theta}(y_t | x, y_{<t}), \quad (1)$$

where $\mathcal{C}$ denotes completion-token positions; prompt tokens are masked from the optimization objective.

²Hugging Face model repository handle.

To mitigate dataset imbalance, we use weighted sampling over groups defined by (base meter, meter subform, length bucket):

$$w(g) \propto \text{count}(g)^{-0.6}.$$

The exponent 0.6 provides moderate rebalancing between unweighted (0) and full inverse-frequency (1) sampling, increasing representation of rare groups without excessive oversampling.

Fine-tuning uses QLoRA (Dettmers et al., 2023), with the base model loaded in 4-bit NF4 precision, double quantization, and bfloat16 computation. LoRA adapters are applied to all linear layers with rank $r = 64$, $\alpha = 128$, dropout 0.05, and RS-LoRA enabled.

Appendix G provides complete training hyperparameters, runtime settings, and instruction templates.

5 Experimental Setup

5.1 Baselines

We evaluate SHAER against three Arabic poetry generation baselines spanning foundation and poetry-specific models:

- **Yehia-7B** (Navid-AI, 2025), the unfine-tuned foundation model underlying SHAER.
- **Ashaar** (Alyafeai et al., 2023), a poetry generation model trained on the Ashaar corpus.
- **Fanar-Diwan** (FANAR Team et al., 2026), a classical Arabic poetry model optimized for prosodic and stylistic constraints.

Yehia-7B provides the cleanest controlled comparison, sharing SHAER’s architecture, tokenizer, and instruction format but lacking task-specific fine-tuning. Ashaar and Fanar-Diwan serve as system-level comparisons under their native conditioning interfaces, which do not support SHAER’s full set of semantic, subform, and length controls. Earlier systems in Section 2 are not directly evaluated due to differing generation and evaluation settings and, where applicable, unavailable public checkpoints or inference implementations.

All models are evaluated on the 3,481-instance held-out test split. Yehia-7B uses the same instruction-style prompt as SHAER, while Ashaar and Fanar-Diwan use their native conditioning formats. Table 2 summarizes conditioning capabilities; Appendices H and I detail input representations and decoding settings, respectively.

5.2 Evaluation Dimensions

Because Arabic poetry quality cannot be captured by a single score, we evaluate generated poems using complementary structural and semantic-literary metrics. Before structural evaluation, outputs are deterministically normalized and segmented into hemistichs and bayts by pairing consecutive hemistichs; any incomplete final hemistich is excluded from bayt-level evaluation. Appendix J provides the full normalization and parsing rules.

Automatic Meter Evaluation. We evaluate meter using an external Arabic meter classifier based on the deep bidirectional recurrent architecture of Abandah et al. (2022). Operating at the verse level, it produces a meter prediction and probability distribution over the supported classes. Appendix K provides complete implementation and evaluation details.

We report two complementary metrics. *Meter Accuracy* measures exact adherence to the requested meter by majority-voting verse-level predictions into a poem-level prediction and computing the percentage matching the target meter.

Meter Conformity provides a softer confidence-based measure of metrical adherence. Let m^* denote the requested meter and $p_i(m^*)$ its predicted probability for verse i . Because meter is a poem-level property, we aggregate verse-level probabilities using the geometric mean:

$$C_{\text{poem}} = \left(\prod_{i=1}^n p_i(m^*) \right)^{1/n}, \quad (2)$$

where n is the number of verses. The geometric mean penalizes metrically inconsistent verses, providing a more faithful estimate of poem-level conformity than an arithmetic average. Final Meter Conformity is the arithmetic mean across all generated poems.

Automatic Meter Subform Evaluation. We evaluate fine-grained prosodic adherence using our reproduced AraPoemBERT 25-class meter-subform classifier (Qarah, 2024), which operates at the bayt level. We report *Bayt-Level Subform Accuracy* and *Poem-Level Subform Accuracy*; the latter averages class-probability vectors across valid bayts and selects the highest-scoring class. The evaluator covers 3,379 of 3,481 test targets (97.07%), excluding targets outside its class inventory. Appendix L details classifier reconstruction, input rep-

Model	Meter	Subform	Rhyme	Hemistich Count	Description
Yehia-7B	✓ [†]	✓ [†]	×	✓ [†]	✓ [†]
Ashaar	✓	×	✓	×	×
Fanar-Diwan	✓	×	✓	×	×
SHAER	✓	✓	×	✓	✓

Table 2: Conditioning capabilities of the evaluated models. [†] denotes variables supplied through prompting rather than learned as native conditioning fields.

resentation, sequence-length configuration, and unsupported meter-subform classes.

For SHAER and Yehia-7B, these metrics measure adherence to the requested subform. Because Ashaar and Fanar-Diwan do not support subform conditioning, their outputs are evaluated against the corresponding source poem’s subform, measuring reference-subform agreement rather than controllability.

Hemistich Count Evaluation. The second structural dimension evaluates adherence to the requested hemistich count using two complementary metrics.

Count Accuracy measures exact count matching as the percentage of generated poems whose hemistich count equals the requested count.

Count Adherence (CA) provides a softer measure of length conformity. Let H_g and $H_r > 0$ denote the generated and requested hemistich counts, respectively:

$$CA = \max \left(0, 1 - \frac{|H_g - H_r|}{H_r} \right). \quad (3)$$

The score equals 1 for an exact match and decreases with relative count discrepancy, reaching 0 when the discrepancy is at least as large as the requested count. Normalization by H_r makes the metric more sensitive to absolute deviations for shorter requested poems. Final Count Adherence is the arithmetic mean across all generated poems.

Both metrics are reported only for models supporting hemistich-count conditioning, namely SHAER and Yehia-7B.

LLM-Based Evaluation. Semantic and literary quality are evaluated using three LLM judges: Qwen3-235B-A22B (Qwen Team, 2025), Qwen 3.7 Max (Qwen Team, 2026b), and GPT-5.6 Terra (OpenAI, 2026). Qwen3-235B-A22B was applied to all 3,481 held-out outputs from each system, while Qwen 3.7 Max and GPT-5.6 Terra evaluate a shared subset of 2,500 test instances across the four systems. Following recent work on LLM-based eval-

Metric	Definition
Description Adherence	Faithfulness to the conditioning description.
Meaning	Clarity and semantic depth of the poem.
Fluency	Linguistic correctness and naturalness of Arabic.
Coherence	Continuity and unity across the poem.
Poeticness	Literary quality beyond meter, including imagery and expression.

Table 3: LLM-based evaluation dimensions.

uation of open-ended generation, we assess five dimensions: *Description Adherence*, *Meaning*, *Fluency*, *Coherence*, and *Poeticness*. Table 3 defines the evaluation dimensions.

All three judges use identical dimension-specific rubrics. Each dimension is assessed independently on a strict 1–5 integer scale, where higher scores indicate better performance. Description Adherence is evaluated only for description-conditioned models, namely SHAER and Yehia-7B. Scores are first averaged over the available valid responses separately for each judge and system. The equal-weight mean of the three judge-level means is reported in the main results. To reduce evaluation variance, all judgments use deterministic decoding (temperature=0). Full evaluation rubrics, decoding settings, coverage details, and judge-specific results are provided in Appendix M.

Human Evaluation. We complement the LLM-based evaluation with a blinded human assessment of top-ranked outputs from all four systems. We evaluate Meaning, Fluency, Coherence, and Poeticness using the same 1–5 rubrics as the LLM judges, excluding Description Adherence because it is not applicable to Ashaar and Fanar-Diwan. For each system, we select the five highest-ranked outputs within the shared instance subset based on aggregate multi-LLM scores, yielding 20 poems. Three authors independently rate the selected poems; all are native Arabic speakers, and one additionally has

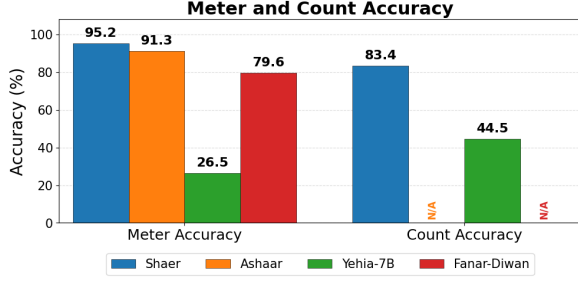


Figure 2: Meter accuracy and count accuracy.

formal training in Arabic prosody and an academic minor in Arabic. Each reported system–dimension score averages 15 ratings across the five poems and three annotators. Further details on selection, blinding, evaluation procedure, and inter-annotator agreement are provided in Appendix N.

Memorization and Copying Evaluation. We additionally assess whether SHAER reproduces content from its training data or from the held-out poems underlying the test instances. Each of the 3,481 generated poems is compared against all 109,070 poems in the SFT training set and against its corresponding held-out source poem after deterministic text normalization.

We measure three complementary indicators of textual overlap: *Exact Match*, which identifies identical normalized poems; *Longest Shared Span*, defined as the longest contiguous sequence of words shared between a generation and a comparison poem; and *5-Gram Jaccard Similarity*, which measures overlap between their sets of word 5-grams. Together, these metrics capture exact reproduction, localized phrase-level reuse, and broader lexical overlap, respectively.

6 Results and Discussion

Appendix O presents representative poems generated by SHAER, together with their conditioning signals and corresponding meter conformity scores. The remainder of this section focuses on quantitative evaluation.

6.1 Formal Control Evaluation

Figures 2 and 3 summarize meter and length-control performance.

SHAER achieves the highest base-meter accuracy (95.2%) and meter-conformity score (0.906), outperforming Ashaar (91.3%, 0.854), Fanar-Diwan (79.6%, 0.649), and Yehia-7B (26.5%, 0.149). It also substantially exceeds Yehia-7B in exact count

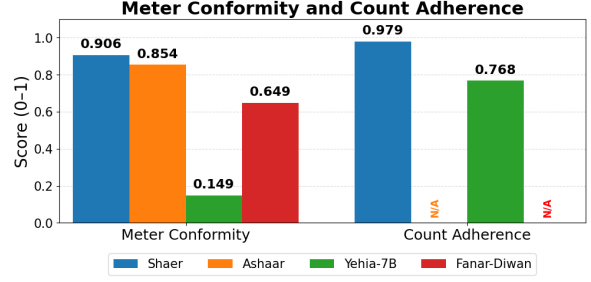


Figure 3: Meter conformity and count adherence scores.

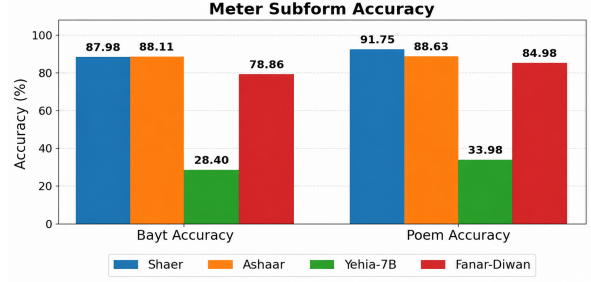


Figure 4: Bayt-level and poem-level meter-subform accuracy.

accuracy (83.4% vs. 44.5%) and count adherence (0.979 vs. 0.768). Because Yehia-7B shares the same foundation model, these differences demonstrate the stronger meter and length control achieved by the jointly conditioned and fine-tuned SHAER system.

Appendix P reports results by base meter and requested-length bucket. SHAER achieves the highest accuracy for 12 of the 13 meters and maintains 93.0–97.0% base-meter accuracy across length buckets. Its exact count accuracy decreases from 98.4% for poems of 1–3 verses to 29.0% for poems of 11–20 verses, but remains substantially above Yehia-7B in every bucket, indicating that longer-form length control remains challenging.

Figure 4 evaluates fine-grained meter-subform control. Among models explicitly conditioned on a target subform, SHAER achieves a bayt-level accuracy of 87.98% and a poem-level accuracy of 91.75%, substantially outperforming Yehia-7B, which obtains bayt-level and poem-level accuracies of 28.40% and 33.98%, respectively. Since both systems share the same foundation model and receive the same target subform, this comparison demonstrates the stronger fine-grained prosodic control achieved by the jointly conditioned and fine-tuned SHAER system.

Since Ashaar and Fanar-Diwan do not support subform conditioning, their scores instead measure

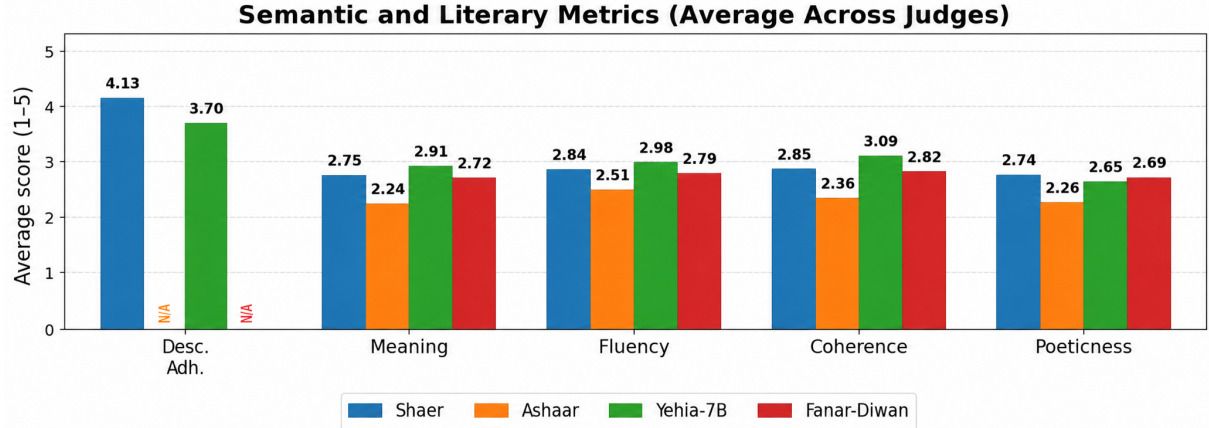


Figure 5: Equal-weight mean semantic and literary scores across three LLM judges. Description adherence applies only to description-conditioned systems.

agreement with the corresponding source poem’s subform. Ashaar achieves 88.11% bayt-level and 88.63% poem-level agreement, while Fanar-Diwan achieves 78.86% and 84.98%, respectively. SHAER attains the highest poem-level accuracy overall, while its bayt-level accuracy is comparable to Ashaar’s reference-subform agreement. These results should be interpreted according to the distinct conditioning settings. Additionally, rare subforms are more frequently classified as the dominant Complete (تام) form; accordingly, evaluation is restricted to target classes represented in the evaluator’s 25-class inventory.

6.2 Semantic and Literary Quality

Figure 5 summarizes semantic and literary quality across the three LLM judges, with judge-specific results reported in Appendix M. Across the four dimensions shared by all systems, SHAER consistently exceeds the two poetry-specific baselines, Ashaar and Fanar-Diwan. The differences relative to Fanar-Diwan are small but consistent across Meaning (2.75 vs. 2.72), Fluency (2.84 vs. 2.79), Coherence (2.85 vs. 2.82), and Poeticness (2.74 vs. 2.69), while Ashaar scores lower across all four dimensions. Since these systems use different native conditioning interfaces, these results represent system-level comparisons rather than controlled comparisons of individual modeling choices.

Compared with its unfine-tuned foundation model, Yehia-7B, SHAER achieves higher cross-judge means for Description Adherence (4.13 vs. 3.70) and Poeticness (2.74 vs. 2.65), whereas Yehia-7B scores higher in Meaning, Fluency, and Coherence. Judge-specific results show the same di-

System	Mean.	Flu.	Coh.	Poet.
SHAER	4.00	4.20	4.20	4.30
Yehia-7B	4.13	3.50	3.80	3.30
Fanar-Diwan	3.80	3.70	3.80	3.50
Ashaar	3.10	3.00	3.60	2.40

Table 4: Blinded human evaluation of five top-ranked poems per system. Scores are means over 15 ratings (five poems $\times$ three annotators) on a 1–5 scale.

rection for Description Adherence, Meaning, Coherence, and Poeticness, while Fluency is judge-dependent. Together with the structural results, these findings indicate that task-specific adaptation substantially improves controllability and description adherence while maintaining competitive semantic and literary quality.

Human Evaluation. Table 4 reports the complementary human evaluation results. SHAER receives the highest ratings for Fluency (4.20), Coherence (4.20), and Poeticness (4.30), while Yehia-7B receives the highest Meaning rating (4.13, compared with 4.00 for SHAER). Fanar-Diwan receives intermediate ratings, whereas Ashaar receives the lowest mean ratings across all four dimensions.

The human assessment partially aligns with the multi-LLM evaluation: both favor SHAER over Yehia-7B on Poeticness and Yehia-7B over SHAER on Meaning, while the human annotators additionally favor SHAER on Fluency and Coherence. Given that this evaluation targets top-ranked outputs rather than a representative test-set sample, these findings provide complementary evidence about the systems’ strongest identified generations rather than their average performance.

6.3 Memorization and Copying Analysis

No exact normalized copies were found among the 3,481 SHAER generations. One generation (0.0287%) shared at least 13 consecutive words with a training poem, and four (0.1149%) with their paired held-out sources; none shared 30 or more words in either comparison. Median word 5-gram Jaccard similarity was 0.0 for both comparisons, with maxima of 0.119 and 0.220, respectively. Manual inspection revealed localized reuse or paraphrasing rather than full-poem reproduction. We therefore find no evidence of systematic memorization or wholesale copying, although rare source-associated phrase reuse occurs.

7 Conclusion

We presented SHAER, a controllable Classical Arabic poetry generation framework jointly conditioned on natural-language descriptions, meter subforms, and target hemistich counts. To support this task, we constructed an enriched corpus of 116,032 classical Arabic poems derived from Ashaar, containing normalized meter-subform labels and automatically generated, validated semantic descriptions. Building on this resource, we adapted Yehia-7B through QLoRA-based supervised fine-tuning with a completion-only objective.

Across the 3,481 held-out test instances, SHAER achieves 95.17% base-meter accuracy, 91.75% poem-level meter-subform accuracy, and 83.40% exact count accuracy, representing substantial gains over its untuned foundation model. It also attains the highest base-meter accuracy among all evaluated systems. Evaluation with three LLM judges indicates that SHAER exceeds the poetry-specific comparison systems across the shared semantic and literary dimensions, while its comparison with Yehia-7B reveals a trade-off between stronger controllability and performance on some unconstrained literary-quality dimensions. A blinded human assessment of top-ranked outputs further supports the model’s competitive literary quality. In addition, memorization analysis finds no exact normalized copies from either the training corpus or the paired held-out source poems.

Together, these findings demonstrate that SHAER provides reliable joint control over semantic intent, fine-grained prosodic form, and poem length while preserving competitive semantic and literary quality. The released code, models, annotations, and evaluation resources provide a foundation for fur-

ther research on controllable Arabic literary generation.

Future work includes explicit rhyme conditioning and evaluation, component-wise ablations, broader human evaluation on representative samples, improved handling of rare meter subforms and longer poems, and extension to additional Arabic poetic genres.

8 Limitations

Despite the strong results achieved by SHAER, several limitations remain. First, this work focuses on Classical Arabic poetry written in the 13 canonical meters retained after corpus filtering. Consequently, the dataset and model do not cover free verse, prose poetry, colloquial poetry, or other contemporary poetic forms. Performance is also less certain for rare meter subforms and longer requested poems because of their limited representation.

Second, 95.6% of the corpus lacked explicit meter-subform metadata and was assigned the canonical Complete (تام) form by default. Although manual inspection and classifier-based validation support this convention, these labels remain partly heuristic.

Third, semantic descriptions are generated automatically rather than written by human annotators. Despite structural validation, corpus-wide quality checks, and human evaluation of sampled descriptions, some residual semantic noise may remain.

Fourth, the evaluation relies partly on learned meter classifiers and LLM judges. Although these evaluators were validated and three LLM judges were used, their scores remain approximations of expert human judgment. The complementary human evaluation is also limited to selected top-ranked outputs rather than a representative test-set sample.

Finally, rhyme is learned implicitly but is neither explicitly controlled nor quantitatively evaluated. Because the experiments do not include component-wise ablations, the observed improvements should be attributed to the jointly conditioned and fine-tuned SHAER system rather than to any individual control mechanism.

References

- Gheith A. Abandah, Mohammed Z. Khedher, Mohammad R. Abdel-Majeed, Hamdi M. Mansour, Salma F. Hulliel, and Lara M. Bisharat. 2022. [Classifying and diacritizing arabic poems using deep recurrent neural](#)

- networks. *Journal of King Saud University - Computer and Information Sciences*, 34(6):3775–3788.
- Asaad Abbas. 2001. The metrical structure of classical arabic poetry. *Studia Arabistyczne i Islamistyczne*, 8:5–52.
- Omar Abboushi and Mohammad Azzeh. 2023. [Toward fluent arabic poem generation based on fine-tuning AraGPT2 transformer](#). *Arabian Journal for Science and Engineering*, 48(8):10537–10549.
- Wafa Al Ghallabi, Ritesh Thawkar, Sara Ghaboura, Ketan Pravin More, Omkar Thawakar, Hisham Cholakkal, Salman Khan, and Rao Muhammad Anwer. 2025. [Fann or flop: A multigenre, multiera benchmark for Arabic poetry understanding in LLMs](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 20224–20244, Suzhou, China. Association for Computational Linguistics.
- Maged S. Al-Shaibani, Zaid Alyafeai, and Irfan Ahmad. 2020. [Meter classification of arabic poems using deep bidirectional recurrent neural networks](#). *Pattern Recognition Letters*, 136:1–7.
- Maytham Alabbas, Zainab A. Khalaf, and Khashan M. Khashan. 2014. [BASRAH: An automatic system to identify the meter of arabic poetry](#). *Natural Language Engineering*, 20(1):131–149.
- Zaid Alyafeai, Maged S. Al-Shaibani, and Moataz Ahmed. 2023. [Ashaar: Automatic analysis and generation of Arabic poetry using deep learning approaches](#). *arXiv*, abs/2307.06218.
- Abd al-Aziz Atiq. 1987. *‘Ilm al-‘Arūd wa-al-Qāfiyah (علم العروض والقافية)*. Dar al-Nahda al-Arabiyya (دار النهضة العربية), Beirut.
- Mohamed El Ghaly Beheitt and Moez Ben Haj Hmida. 2022. [Automatic arabic poem generation with GPT-2](#). In *Proceedings of the 14th International Conference on Agents and Artificial Intelligence*, volume 2, pages 366–374. SciTePress.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient finetuning of quantized LLMs](#). In *Advances in Neural Information Processing Systems*, volume 36.
- FANAR Team, Ummar Abbas, Mohammad Shahmeer Ahmad, Minhaj Ahmad, Abdulaziz Al-Homaid, Anas Al-Nuaimi, Enes Altinisik, Ehsaneddin Asgari, Sanjay Chawla, Shammur Chowdhury, Fahim Dalvi, Kareem Darwish, Nadir Durrani, Mohamed Elfeky, Ahmed Elmagarmid, Mohamed Eltabakh, Asim Ersoy, Masoomali Fatehkia, Mohammed Qusay Hashim, and 18 others. 2026. [Fanar 2.0: Arabic generative AI stack](#). *arXiv*, abs/2603.16397.
- Navid-AI. 2025. [Yehia 7b preview](#).
- OpenAI. 2026. [Gpt-5.6 terra](#). OpenAI API Documentation. Accessed: 2026-09-16.
- Faisal Qarah. 2024. [AraPoemBERT: A pretrained language model for Arabic poetry analysis](#). *arXiv*, abs/2403.12392.
- Qwen Team. 2025. [Qwen3 technical report](#). *arXiv*.
- Qwen Team. 2026a. [Qwen3.5: Towards native multi-modal agents](#). Accessed: 2026-04-27.
- Qwen Team. 2026b. [Qwen3.7](#). Qwen Blog. Accessed: 2026-09-16.
- Abdelrahman Sadallah, Kareem Elozeiri, Mervat Abassy, Rania Elbadry, Mohamed Anwar, Abed Alhakim Freihat, Preslav Nakov, and Fajri Koto. 2026. [Instruction-guided poetry generation in Arabic and its dialects](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 38761–38780, San Diego, California, United States. Association for Computational Linguistics.
- Sameerah Talafha and Banafsheh Rekabdar. 2019. [Arabic poem generation with hierarchical recurrent attentional network](#). In *2019 IEEE 13th International Conference on Semantic Computing (ICSC)*, pages 316–323. IEEE.
- Sameerah Talafha and Banafsheh Rekabdar. 2021. [Poetry generation model via deep learning incorporating extended phonetic and semantic embeddings](#). In *2021 IEEE 15th International Conference on Semantic Computing (ICSC)*, pages 48–55. IEEE.
- Mohamad Bilal Zbib, Hasan Abed Al Kader Hammoud, Ammar Mohanna, Nadine Rizk, Fatima Karnib, Sina Moukaled, and Bernard Ghanem. 2026. [Araling-bench: A human-annotated benchmark for evaluating arabic linguistic capabilities of large language models](#). In *Proceedings of the 2nd Workshop on NLP for Languages Using Arabic Script*, pages 385–393, Rabat, Morocco. Association for Computational Linguistics.

Appendix

A Background on Classical Arabic Poetry

A.1 Poetic Structure

Classical Arabic poetry follows a highly structured form governed by meter and rhyme. A poem (قصيدة, *qasida*) consists of verses (أبيات, *abyat*; singular بيت, *bayt*), each typically divided into two hemistichs (شطرين, *shatrayn*): الصدر (*sadr*, first hemistich) and العجز (*ajuz*, second hemistich). Hemistichs within a poem typically conform to a single meter (بحر, *bahr*), while the verses share a common rhyme (قافية, *qafiyah*), realized at the end of the second hemistich (Abbas, 2001). These constraints require poetry generation to maintain both semantic and prosodic consistency.

English Name	Arabic Name	Meter Key
Taweel	الطويل	فعولن مفاعيلن فعولن مفاعيلن
Madeed	المديد	فاعلاتن فاعلن فاعلاتن
Baseet	البسيط	مستفعلن فاعلن مستفعلن فاعلن
Wafer	الوافر	مفاعلاتن مفاعلاتن فعولن
Kamel	الكامل	مفاعلاتن مفاعلاتن مفاعلاتن
Hazaj	الهزج	مفاعيلن مفاعيلن
Ramal	الرمل	فاعلاتن فاعلاتن فاعلاتن
Rajaz	الرجز	مستفعلن مستفعلن مستفعلن
Sare'e	السريع	مستفعلن مستفعلن فاعلن
Munsareh	المنسرح	مستفعلن مفعولات مستفعلن
Khafeef	الخفيف	فاعلاتن مستفعلن فاعلاتن
Mujtath	المجث	مستفعلن فاعلاتن
Mutakareb	المتقارب	فعولن فعولن فعولن فعولن
Mutadarak	المتدارك	فاعلن فاعلن فاعلن فاعلن
Muqtadheb	المقتضب	فاعلات مفتعلن
Mudhare'e	المضارع	مفاعيلن فاعلاتن

Table 5: The sixteen canonical Khalilian meters and their corresponding meter keys (Alabbas et al., 2014).

A.2 Meter and Meter Subforms

Arabic prosody (علم العروض, *ilm al-ʿarud*) describes the metrical organization of classical Arabic poetry. Each meter (بحر, *bahr*) is characterized by an ordered sequence of recurring metrical feet (تفاعيل, *tafaʿil*) (Abbas, 2001). For example, the canonical pattern of Rajaz (الرجز) is مستفعلن مستفعلن in each hemistich. Table 5 summarizes the sixteen canonical Khalilian meters and their meter keys.

A base meter can admit multiple valid realizations through systematic modifications of its canonical structure (Atiq, 1987). We refer to these finer-grained realizations as *meter subforms*. Common examples include Complete (تام), Catalectic (مجزوء), Hemistich (مشطور), and Severely Truncated (منهوك); these are illustrative rather than exhaustive, and valid subforms vary across meters. Consequently, two poems can share the same base meter while following different subforms. Table 6 illustrates this distinction using Rajaz.

B Analysis of the Original Ashaar Corpus

The original Ashaar corpus contains 254,630 poems and 12 metadata fields per poem. However, substantial inconsistencies in metadata coverage, meter annotations, description availability, poem length, and textual quality motivated the cleaning and filtering procedures described in Section 3. Table 7 summarizes the key statistics.

C Data Cleaning and Preprocessing

We apply a four-stage pipeline to remove structural, orthographic, and metadata inconsistencies from the original Ashaar corpus. First, the corpus is restricted to classical Arabic poetry written in the sixteen canonical Khalilian meters. Poems labeled as colloquial or non-classical (عامي, شعبي), as well as poems lacking meter annotations, are discarded. Second, poems with malformed verse structures, empty hemistichs, or hemistichs containing Latin characters, digits, or URLs are removed. Internal elongation (تطويل (-)) characters are also stripped, while edge elongation and all diacritics are preserved. Third, poems exceeding twenty verses are excluded to align the corpus with the supervised fine-tuning maximum sequence length of 1,024 tokens. A post-tokenization audit using the final instruction format found that only 4 of the 109,070 training sequences (0.0037%) still exceeded this limit, by 5–25 tokens, while no validation or test sequence exceeded it. Finally, meter annotations are normalized to the canonical meter inventory, and base meters represented by fewer than 500 poems are removed. Figure 6 summarizes the resulting counts at each stage.

D Meter Subform Details

D.1 Default-Label Validation

We first manually assessed the default Complete assignment on a sample of 1,000 poems. An author with formal training in Arabic prosody inspected the sampled poems and confirmed the Complete

Name	# Feet	Metrical Feet Pattern	Definition	Example
Complete تام	6	مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ	Retains the full three-foot pattern in each of the two hemistichs.	لَا خَيْرَ فِيمَنْ كَفَّ عَنْهُ شَرُّهُ إِنْ كَانَ لَا يَرْجَى لِيَوْمٍ خَيْرُهُ
Truncated مُجْزِئ	4	مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ	Retains two feet in each of the two hemistichs.	فَارْفُقْ بِصَبِّ ذِي جَوَى مَنْ حَرَّهُ ذَابَ الْحَشَى
Single-Hemistich مُشْطُور	3	مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ	Consists of a single three-foot verse.	إِنْ أَبْتَيْ بِبِضَاءٍ مَنْ بِيضٍ زَهْرُ يَا لَيْتَنِي فِيهَا جَدَعُ
Exhausted مَنْهُوك	2	مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ مُسْتَفْعِلُنْ	Consists of a single two-foot verse.	

Table 6: Illustrative meter subforms of Rajaz (بحر الرجز), showing the number of metrical feet per verse, their metrical patterns, and corresponding examples. The set is not exhaustive.

Statistic	Value
Total number of poems	254,630
Number of metadata fields	12
Number of unique meter labels	101
Poems lacking meter annotations	39.77%
Poems containing at least one empty hemistich	0.17%
Number of unique theme categories	19
Poems lacking theme annotations	73.48%
Poems with unreliable language-type labels (عامي, شعبي, missing, or “-”)	31.44%
Poems lacking descriptions	94.32%
Average number of hemistichs per poem	30.30
Median number of hemistichs per poem	14
Minimum hemistich count	0
Maximum hemistich count	11,608
Poems exceeding 40 hemistichs (20 verses)	21.68%
Poems containing Latin characters, numerical digits, or URLs	2.56%
Poems containing <i>tatweel</i> characters	45.45%

Table 7: Key characteristics of the original Ashaar corpus before preprocessing and filtering.

form in 97.2% of cases. This manual check evaluates the plausibility of the defaulting convention on a sample; it does not constitute exhaustive re-annotation of the 110,926 default-labeled poems.

We reproduce the meter-subform classification setting of AraPoemBERT (Qarah, 2024) using the dataset released by the authors, as the publicly released checkpoint could not be directly loaded as a complete sequence-classification model. Our reproduction achieved 97.38% accuracy, comparable to the 97.79% reported in the original work. We apply the resulting classifier to all poems assigned the default Complete label. Table 8 reports the resulting agreement by base meter.

The four meters with low agreement reflect limitations of the classifier itself. Complete forms of Madeed, Hazaj, and Mujtath are absent from the classifier’s released class inventory, while Rajaz Complete has a reported precision of 0.4725 (Qarah, 2024). In contrast, agreement ranges from

98.76% to 99.93% across the remaining nine meters. We use these predictions only to assess the default-labeling strategy and do not use them to modify the corpus annotations.

D.2 Meter–Subform Distribution

Table 9 reports the distribution of the 21 meter–subform combinations in the final corpus.

E Semantic Description Generation and Validation

The description-generation pipeline consists of three components: selection of the generation configuration, specification of the prompting and decoding procedure for the selected configuration, and validation of the resulting corpus descriptions.

E.1 Generation Configuration Selection

Before generating descriptions for the full corpus, we compare three candidate generation configura-

Base Meter	Default Complete	Predicted Complete	Predicted Non-Complete	Agreement
Taweel	28,028	28,007	21	99.93%
Mutakareb	5,189	5,185	4	99.92%
Baseet	16,811	16,787	24	99.86%
Kamel	19,438	19,399	39	99.80%
Khafeef	9,555	9,531	24	99.75%
Wafer	11,097	11,047	50	99.55%
Sare'e	7,190	7,154	36	99.50%
Ramal	3,643	3,615	28	99.23%
Munsareh	2,747	2,713	34	98.76%
Madeed	538	37	501	6.88%
Rajaz	3,973	225	3,748	5.66%
Hazaj	609	15	594	2.46%
Mujtath	2,060	33	2,027	1.60%
Overall	110,926	103,748	7,178	93.53%

Table 8: Agreement between the default Complete annotations and predictions of the reproduced AraPoemBERT meter-subform classifier, reported by base meter.

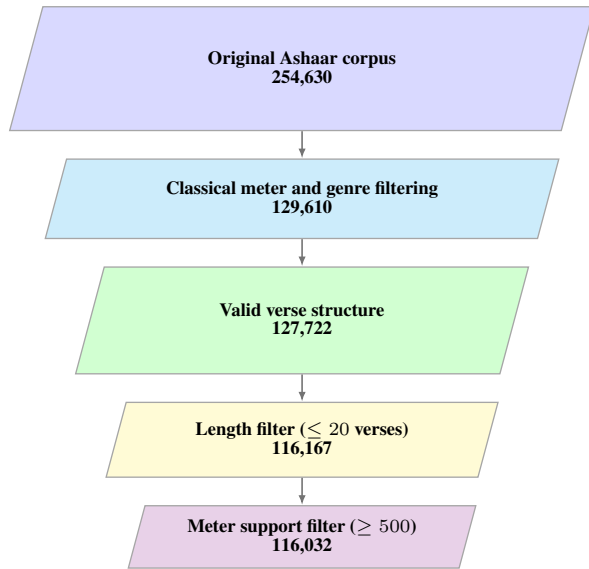


Figure 6: Corpus filtering pipeline. Starting from the original Ashaar corpus, the filtering process yields a final dataset of 116,032 classical Arabic poems.

Base Meter	Subform	Poems
Baseet	Complete	16,811
Baseet	Mukhalla'	651
Baseet	Majzu'	159
Khafeef	Complete	9,555
Khafeef	Majzu'	298
Rajaz	Complete	3,973
Rajaz	Majzu'	612
Ramal	Complete	3,643
Ramal	Majzu'	949
Sare'e	Complete	7,190
Taweel	Complete	28,028
Kamel	Complete	19,438
Kamel	Majzu'	1,862
Kamel	Ahadh	373
Mutakareb	Complete	5,189
Mujtath	Complete	2,060
Madeed	Complete	538
Munsareh	Complete	2,747
Hazaj	Complete	609
Wafer	Complete	11,097
Wafer	Majzu'	250

Table 9: Distribution of meter-subform combinations in the final corpus.

tions differing in generative model and prompting strategy. For each configuration, we generate descriptions for a stratified sample of 600 poems covering diverse base meters, meter subforms, and poem lengths.

Each generated description is independently assigned a Pass or Fail label by three of the authors, who are blinded to the configuration that produced it. All three annotators are native Arabic speakers; one additionally has formal training in Arabic prosody. Descriptions are assessed according to nine criteria: fluency, faithfulness to the poem content, coverage of the central meaning and imagery, absence of hallucinated content, absence of direct verse copying, sufficient specificity, use of the

required opening phrase (القصيدة تتحدث عن...), single-paragraph formatting, and absence of meta-data leakage. A description receives a Fail label if any criterion is violated.

Disagreements are resolved by majority vote. Inter-annotator agreement across the evaluations is Fleiss' $\kappa = 0.817$, indicating strong consistency in the application of the evaluation rubric.

The three candidate configurations achieve pass rates of 428/600 (71.3%), 506/600 (84.3%), and 557/600 (92.8%), respectively. We select the third configuration, which achieves the highest pass rate, for corpus-wide description generation.

Parameter	Value
Model	Qwen/Qwen3.5-35B-A3B
Temperature	0.0
Maximum output tokens	300
Response format	JSON object
Reasoning effort	Disabled
Provider fallbacks	Disabled

Table 10: Decoding configuration used for corpus-wide semantic description generation.

E.2 Prompting and Decoding Configuration

The selected configuration uses Qwen3.5-35B-A3B (Qwen Team, 2026a) under deterministic decoding. The system prompt instructs the model to produce a single descriptive Arabic paragraph beginning with (القصيدة تتحدث عن...), capturing the poem’s central meaning, imagery, and tone. It further instructs the model to avoid copying verses, introducing unsupported content, or referring to formal properties such as meter, rhyme, or poem length. The response is returned as a JSON object containing a single new_description field.

The user prompt provides all hemistichs of the source poem followed by a one-line instruction requesting the semantic description.

الآيات:

[All poem hemistichs]

اكتب وصفاً واحداً يلتقط أفكار القصيدة ومعناها
وصورها ونبرتها العامة.

The decoding configuration used for corpus-wide generation is reported in Table 10.

E.3 Description Validation

Following configuration selection, descriptions generated for the full corpus undergo a two-stage validation procedure. The first stage ensures structural validity during generation, while the second performs fine-grained poem-level quality checks over the generated descriptions. Descriptions that fail verification are regenerated and re-evaluated before inclusion in the final corpus.

E.3.1 Stage 1: Automatic Structural Validation

Each description is validated immediately after generation. The pipeline parses the JSON response, verifies the presence of the new_description field,

normalizes whitespace, and rejects empty, malformed, or schema-leaking outputs. Failed generations are retried up to four times. Publication is blocked if any poem is missing a valid description. This stage ensures structural correctness and does not assess semantic quality.

E.3.2 Stage 2: Corpus-Wide Fine-Grained Validation

We subsequently perform deterministic poem-level quality checks over the complete set of generated poem-description pairs. These checks assess adherence to the required output format, leakage of formal poem metadata, and direct copying from the source poem. Table 11 reports first-pass results before regeneration. Descriptions failing one or more checks were regenerated and re-evaluated, and all descriptions retained in the final corpus passed the Stage 2 checks.

Format-compliance checks verify that descriptions are non-empty, begin with the required opening phrase, consist of a single paragraph, are not formatted as lists, do not take the form of questions, requests, or commands, and contain no leaked JSON syntax. Metadata-leakage checks identify references to formal poetic properties that descriptions were instructed not to mention. Copying checks detect full hemistich reproduction, long copied phrases, and excessive overlap with the source poem.

F Final Dataset Characteristics

The released artifacts distinguish between newly created resources and inherited corpus content. Our code, adapters, and separable annotations and semantic descriptions are released under the Apache-2.0 license. The underlying poem text remains subject to Ashaar’s non-commercial and research-use restrictions and reserved takedown rights; our release does not relicense that content.

Table 12 summarizes the principal characteristics of the final dataset.

G Supervised Fine-Tuning Details

Table 13 summarizes the training hyperparameters and runtime settings used during QLoRA fine-tuning, and the instruction templates used during dataset construction are provided below.

G.1 Instruction Templates

Training examples are constructed using a fixed instruction format. The system prompt defines the

Category	Metric	Pass Rate
Format Compliance	Non-empty description	100%
	Required opening phrase	100%
	Single paragraph	100%
	Not a list	100%
	Not a question/request/command	99.29%
	No JSON leakage	99.99%
Metadata Leakage	No metadata leakage	98.34%
Copying Detection	No full hemistich copied	97.46%
	No long phrase copied	97.43%
	No excessive copying	99.49%

Table 11: First-pass corpus-level validation results for automatically generated semantic descriptions, reported before regeneration. Pass rates measure compliance with formatting requirements, absence of metadata leakage, and avoidance of direct copying from source poems. Failed descriptions were regenerated and re-evaluated before inclusion in the final corpus.

Statistic	Value
Poems	116,032
Base meters	13
Meter-subform pairs	21
Average verses per poem	5.52
Median verses per poem	4
Generated descriptions	116,032
Average description length	266.2 chars

Table 12: Summary statistics of the final SHAER dataset after preprocessing, filtering, meter subform annotation, and semantic description generation.

model’s role as a classical Arabic poet and enforces strict meter adherence, while the user prompt supplies the target meter, meter subform, hemistich count, and semantic description.

G.1.1 System Prompt

أنت شاعر عربي تكتب الشعر العمودي
الكلاسيكي.
التزم بالبحر المحدد في كل شطر، واستلهم
من الموضوع دون نقله حرفياً.
أخرج الأبيات فقط دون مقدمة أو تعليق.
التزم التزاماً صارماً بالبحر المطلوب، ولا
تخرج عنه.

G.1.2 User Prompt Template

{base_meter} : البحر الأساسي:
{form} : الصيغة:
{num_lines} : عدد الأشر المطلوب:
{description} : الموضوع:

{form} اكتب {num_lines} أشطراً ملتزمة بصيغة {form}
من بحر {base_meter} دون أي شرح إضافي.

During training, placeholders are replaced with the corresponding poem metadata and semantic description. The loss is computed only over the completion tokens.

H Native Input Representation

Table 14 summarizes the native input representations and conditioning signals used by the evaluated models.

I Poem Generation Decoding

Table 15 reports the decoding and stopping settings used to generate the evaluated poems.

J Generation Parsing Details

Structural evaluation uses the generated_text field for every system. We normalize line endings, remove invisible directional and control characters, trim surrounding whitespace, and remove empty lines. A line that already contains a recognized two-hemistich separator is treated as a complete bayt candidate. Otherwise, consecutive non-empty lines are paired in generation order: lines 1–2 form bayt 1, lines 3–4 form bayt 2, and so forth. Pairing is independent of the requested poem length.

With REQUIRE_TWO_SHATR=True, only complete two-hemistich bayts are classified. An unmatched final hemistich is retained only for coverage and audit accounting and is excluded from bayt-level structural classification.

K Meter Classifier

The meter classifier follows the four-layer Bidirectional Long Short-Term Memory (BiLSTM) architecture of Abandah et al. (2022), shown in Figure 7.

Parameter	Value
Base model	Navid-AI/Yehia-7B-preview
Training objective	Causal language modeling with completion-only loss
Prompt loss masking	Enabled; prompt tokens are masked with -100
Max sequence length	1024
Train packing	Enabled
Truncation mode	keep_end
Tokenizer padding side	Left
Quantization	QLoRA, 4-bit NF4, double quantization, bfloat16
LoRA target modules	all-linear
LoRA rank / alpha / dropout	64 / 128 / 0.05
LoRA bias	none
LoRA task type	CAUSAL_LM
RS-LoRA	Enabled
Optimizer	paged_adamw_8bit
Learning rate	8×10^{-5}
Scheduler	Cosine
Warmup ratio	0.03
Weight decay	0.0
Max gradient norm	1.0
Batch size per device	1
Gradient accumulation	16
Effective batch size	16
Epochs	2
Global steps	4,236
Evaluation interval	200 steps
Save interval	200 steps
Best-model metric	eval_loss
Seed	42
Train/val/test split	94 / 3 / 3 (stratified by base meter, meter subform, and length bucket)
Length buckets	1 to 3, 4 to 6, 7 to 10, and 11 to 20 verses
Training sampler	Weighted sampler with $w(g) \propto (1/\text{count}(g))^{0.6}$ over (base meter, meter subform, length bucket)
Runtime	12.53 hours

Table 13: Training hyperparameters, optimization settings, and runtime configuration used for QLoRA-based supervised fine-tuning of SHAER.

Each input verse is represented as a fixed-length character sequence ($T = 128$) over a vocabulary of 46 characters, mapped into dense embeddings and processed by four stacked BiLSTM layers before a classification head predicts one of 17 classes corresponding to the supported meter labels and prose. Our PyTorch implementation is trained and evaluated on the APCD_plus_prose_all dataset, achieving **97.24%** accuracy on the held-out test split of 163,917 samples. The trained classifier is released as part of the project resources and is used for all metrical evaluations reported in this work.

L Meter-Subform Classifier

Because the publicly available sub-meter checkpoint did not contain a complete trained sequence-classification head, we reconstructed the released AraPoemBERT sub-meter task (Qarah, 2024) and trained an ARBERT-based classifier using the published configuration. Each complete bayt is represented as first_hemistich S second_hemistich,

with a maximum sequence length of 32 tokens.

The 102 unsupported test targets comprise Baseet–Majzu’ (6 poems), Hazaj–Complete (18), Madeed–Complete (17), and Mujtath–Complete (61); Kamel–Ahadh is represented in the evaluator’s class inventory.

M LLM-Based Evaluation

Tables 16 and 17 summarize the judge models and decoding configurations, while Table 18 reports the complete judge-specific results.

Each evaluation dimension is submitted through a separate request for every poem. The expected response is a forced function call containing an integer score from 1 to 5. Scores are parsed first from the function-call arguments. If these are unavailable, the parser accepts JSON content, fenced or embedded JSON, or a standalone integer from 1 to 5.

Malformed responses, parsing failures, and HTTP errors are retried up to three times after the

Model	Native Input Format	Conditioning Signals	Output
Yehia-7B (Navid-AI, 2025)	Instruction-style prompt using the same format as SHAER, but without task-specific fine-tuning.	Meter, meter subform, hemistich count, and semantic description provided through prompting only.	Generated poem
Ashaar (Alyafeai et al., 2023)	Structured HTML-like representation using special tokens for meter, theme, qafiyah, poem separators, bayt separators, and verse separators.	Base meter, theme, and rhyme/qafiyah.	Generated or completed poem
Fanar-Diwan (FANAR Team et al., 2026)	Metadata-conditioned prompt consisting of a hemistich followed by bracketed poetic metadata, such as meter, topic, era, poet identifier, and rhyme letter.	Base meter, topic, era, poet identity, and rhyme letter.	Poem continuation
SHAER	Arabic instruction-following prompt explicitly specifying the target meter, meter subform, number of hemistichs, and semantic description.	Meter, meter subform, hemistich count, and semantic description.	Controlled generated poem

Table 14: Native input representations, conditioning signals, and output formats of the evaluated models.

System	Temperature	Top- p	Top- k	Maximum new tokens	Repetition penalty	Stopping and generation-time post-processing
SHAER	0.55	0.90	Not set	768	1.08	Model EOS or the 768-token generation limit; no custom stopping strings or token IDs.
Yehia-7B	0.55	0.90	Not set	768	1.08	Model EOS or the 768-token generation limit; no custom stopping strings or token IDs.
Fanar-Diwan	0.55	0.90	Not set	768	1.08	Model EOS or the 768-token generation limit; no custom stopping conditions.
Ashaar	0.55	0.90	Not set	512	1.08	Model EOS or the 512-token generation limit. Post-processing also truncates the output when decoded control tokens indicating meter or theme are encountered.

Table 15: Poem-generation decoding settings used for the evaluated systems. Top- k was not explicitly supplied and therefore followed the relevant inference framework or model-configuration default.

Parameter	Value
Judge model	qwen/ qwen3-235b-a22b-2507
Temperature	0.0
Max output tokens	128
Reasoning effort	None
Tool calling	Forced
Output format	Integer score (1 to 5)

Table 16: Decoding configuration used for Qwen3-235B-A22B.

initial attempt, allowing at most four attempts, with delays of 2, 4, and 6 seconds. The request timeout is 120 seconds, followed by a 0.25-second post-request interval. A permanently failed request terminates the run rather than silently excluding the corresponding evaluation. The completed runs contain no unresolved errors.

Qwen3-235B-A22B was applied to all 3,481 generated poems from each system, and its reported means are computed over the available valid scores. Qwen 3.7 Max and GPT-5.6 Terra evaluate the corresponding outputs from all four systems on a shared subset of 2,500 test instances, with complete coverage for every applicable system–dimension combination. SHAER and Yehia-7B are evaluated on all five dimensions, whereas Ashaar and Fanar-Diwan are evaluated on the four shared literary dimensions because their native interfaces do not support description conditioning.

For the cross-judge summary reported in the main text, scores are first averaged over the evaluated instances separately for each judge and system. The resulting three judge-level means are then averaged with equal weight, ensuring that each judge contributes equally despite the different evaluation-

Setting	Qwen 3.7 Max	GPT-5.6 Terra
Model/API identifier	qwen/qwen3.7-max	openai/gpt-5.6-terra
API provider	OpenRouter Chat Completions	OpenRouter Chat Completions
Temperature	0	0
Top- p / top- k	Provider defaults; fields not sent	Provider defaults; fields not sent
Maximum output tokens	128	128
Reasoning effort	None	None
Seed	Provider default; no seed sent	Provider default; no seed sent
Structured output	Forced submit__score function call	Forced submit__score function call
Client-side fallback model	None	None

Table 17: API and decoding configurations used for Qwen 3.7 Max and GPT-5.6 Terra.

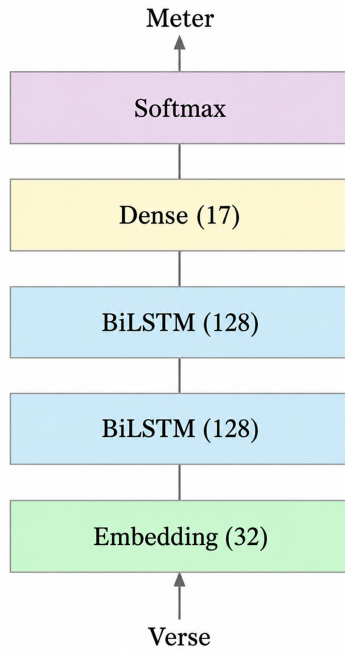


Figure 7: Architecture of the 4-BiLSTM Arabic meter classifier used for automatic meter evaluation.

set sizes. All three judges use identical dimension-specific evaluation rubrics, provided below.

M.1 Evaluation Rubrics

M.1.1 Description Adherence

- 1: Off-topic, contradictory, or ignores the central request.
- 2: Touches only the broad topic while missing the intended meaning.
- 3: Follows the main topic but only partially realizes the requested details.
- 4: Clearly follows the description with only minor omissions.

- 5: Precisely realizes the requested description and appears specifically written for that prompt.

M.1.2 Meaning

- 1: No recoverable meaning or severe contradictions.
- 2: Weak meaning with little semantic substance.
- 3: Understandable but ordinary or weakly developed meaning.
- 4: Clear and worthwhile meaning with genuine semantic content.
- 5: Rich and non-trivial poetic meaning exhibiting substantial depth or insight.

M.1.3 Fluency

- 1: Severely corrupted or ungrammatical Arabic.
- 2: Frequent errors that substantially reduce readability.
- 3: Generally understandable but containing noticeable awkwardness or errors.
- 4: Mostly fluent with only minor imperfections.
- 5: Consistently natural, correct, and elegant classical-style Arabic.

M.1.4 Coherence

- 1: Fragmented or internally inconsistent.
- 2: Frequent topic shifts and weak overall unity.
- 3: General thematic consistency but limited development.

System	Judge	Description	Meaning	Fluency	Coherence	Poeticness
SHAER	Qwen3-235B-A22B	4.75	3.72	4.31	3.69	3.82
Yehia-7B	Qwen3-235B-A22B	4.42	3.75	4.21	3.82	3.73
Fanar-Diwan	Qwen3-235B-A22B	–	3.70	4.26	3.69	3.79
Ashaar	Qwen3-235B-A22B	–	2.84	3.46	2.84	3.00
SHAER	Qwen 3.7 Max	3.88	2.35	2.07	2.37	2.19
Yehia-7B	Qwen 3.7 Max	3.39	2.56	2.23	2.64	2.13
Fanar-Diwan	Qwen 3.7 Max	–	2.33	2.06	2.35	2.17
Ashaar	Qwen 3.7 Max	–	2.05	2.01	2.18	1.98
SHAER	GPT-5.6 Terra	3.75	2.19	2.15	2.49	2.21
Yehia-7B	GPT-5.6 Terra	3.30	2.43	2.50	2.81	2.09
Fanar-Diwan	GPT-5.6 Terra	–	2.13	2.05	2.41	2.12
Ashaar	GPT-5.6 Terra	–	1.82	2.06	2.08	1.80

Table 18: Mean semantic and literary evaluation scores reported separately for each LLM judge. Qwen3-235B-A22B was applied to all 3,481 outputs per system, while Qwen 3.7 Max and GPT-5.6 Terra evaluate a shared subset of 2,500 test instances. Means are computed over the available valid scores. Description adherence applies only to SHAER and Yehia-7B; dashes indicate inapplicable evaluations.

- 4: Mostly unified with clear line-to-line continuity.
- 5: Strongly coherent with meaningful progression of ideas, imagery, and emotion.

M.1.5 Poeticness

- 1: Flat, mechanical, prose-like, or non-poetic output.
- 2: Weak poetic quality with limited imagery or emotional impact.
- 3: Some poetic characteristics but limited artistic development.
- 4: Clearly poetic with effective literary qualities.
- 5: Exceptional poetic quality with vivid imagery, strong emotional resonance, elegant rhetoric, and memorable diction.

N Human Evaluation

N.1 Selection and Evaluation Procedure

For ranking, the aggregate score of each candidate output is computed as the equal-weight mean across the three LLM judges and the four applicable evaluation dimensions. Selection is performed independently within each system.

System identities are removed, and the selected poems are presented in randomized order to reduce model-identity and presentation-order biases. Each annotator independently assigns one score per applicable dimension using the rubrics defined in appendix M. Meter, meter-subform adherence, and

length control are not assessed, as the human evaluation focuses exclusively on semantic and literary quality.

N.2 Inter-Annotator Agreement

We assess inter-annotator agreement separately for each dimension using ordinal Krippendorff’s α , accounting for the ordered 1–5 rating scale and the presence of three annotators.

Agreement is highest for Coherence ($\alpha = 0.901$), while Meaning ($\alpha = 0.758$), Fluency ($\alpha = 0.751$), and Poeticness ($\alpha = 0.756$) exhibit lower but similar agreement.

O Qualitative Examples from SHAER

Table 19 presents selected qualitative examples generated by SHAER. For each example, we report the input conditioning signals and the resulting meter conformity score.

P Performance Breakdowns

Tables 20 and 21 report structural performance by base meter and requested-length bucket, respectively. All 3,481 held-out instances are included for every system; no rows are excluded from these breakdowns.

Meter	Form	Bayts	Description and Generated Poem	Meter score
البسيط	مخلع	3	Description: القصيدة تتحدث عن شخص ظهرت عليه ملامح الحسن ، حيث تكشف ملامح وجهه عن فنون من الظرف والجمال، لدرجة أن وصله أصبح حقيقة يقينية تفوق ما كان في الخيال. Generated poem: قد لاح لي شخصه فأبدى للفضل في حسنه فنون في وجهه طالعات حسن وفي طلي طرفه فنون قد صار وصلي به يقينا ما كان قبل الخيال ظن	0.9886
الكامل	أخذ	2	Description: القصيدة تتحدث عن لحظة ظهور عارضٍ لامعٍ في وجه الحبيب، يُقارن بجمال الرياح والآس، مما يدفع المتكلم إلى تقبيله فرحاً، لكن هذا التقبيل يتحول فجأة إلى سواد ناتج عن نيران أنفاسه، مما يعكس تبايناً بين الفرح والحرقه في مشهد عاطفي مكثف. Generated poem: لاح العذارُ بخده فكانه رِيحانةٌ وأس قبلته طرباً وهواً فغدا السوادُ به هو المقياسُ	0.9859
الرملي	مجزوء	2	Description: القصيدة تتحدث عن تأثير الشيخوخة الذي قوس القناتين بعد أن كاتنا سويتين، وتصور الشعر الأبيض الذي يهتز برأس الشاعر وكأنه نشيد يهتف به الموت، مما يجعله عنواناً للنية. Generated poem: قوس الشيخُ قناتي بعدما كانت سويه وإذا أبصرت شبي هتفت بي للنية	0.9941
المجتث	تام	2	Description: القصيدة تتحدث عن مفارقة كأس المحبة الذي يجمع بين المرارة والحلاوة، لتعبر عن طبيعة الحب الذي يبدو للوهلة الأولى بعيداً لكنه في الحقيقة قريب جداً من القلب. Generated poem: كأس المحبة فيها مرارة وحلاوة والحب من بعد قرب للقلب فيه غواية	0.9627
الرجز	تام	2	Description: القصيدة تتحدث عن موقف الشجاعة والصدق في الخفاء، حيث يصف الشاعر نفسه بأنه يحمل ما حُل من خير وشر، ويؤكد أن صدق قناته لا يتشبه بوجه العذر الكاذب، مستخدماً صورة الشهاب كالشر في الكف لتدل على القوة الكامنة التي تظهر حين يختفي الجبان. Generated poem: إني إذا غبتَ وحملتُ الخبرَ حملتُ ما حُمِلتُ من خيرٍ وشرٍ صدقُ قناتي ليس كذباً كوهج كأنها شهبٌ بكفٍّ من شرٍ	0.9968
البسيط	مخلع	2	Description: القصيدة تتحدث عن دعاء المولى أن يكون الله مفرجاً عن صدر الشاعر بعد أن طال به ضيقه واكتابه، مستنداً في ذلك إلى فكرة أن اتساع الصدر يترافق مع النعم، بينما ضيقه يترافق مع العذاب. Generated poem: يا ربِّ كن لي مُفرِّجاً ما طال بي الضيقُ والكابُ فالصدرُ بالنعماءِ واسعٌ والصدرُ بالنعماتِ ضائقُ	0.6597

Table 19: Qualitative examples generated by SHAER. For each example, the table reports the conditioning signals used for generation, the generated poem displayed in bayt structure, and its meter conformity score.

Base Meter	N	SHAER	Yehia-7B	Ashaar	Fanar-Diwan
Baseet	529	98.49	24.76	95.27	79.21
Khafeef	295	96.61	10.17	97.97	83.73
Rajaz	137	80.29	15.33	64.23	58.39
Ramal	138	84.06	9.42	83.33	63.04
Sare'e	216	92.13	6.94	78.70	64.81
Taweel	841	99.88	61.00	97.62	94.89
Kamel	649	95.53	18.03	92.45	79.20
Mutakareb	156	96.79	4.49	91.03	79.49
Mujtath	61	83.61	6.56	73.77	63.93
Madeed	17	64.71	5.88	41.18	29.41
Munsareh	84	82.14	0.00	75.00	48.81
Hazaj	18	83.33	0.00	55.56	50.00
Wafer	340	95.59	20.59	95.29	78.82
Overall	3,481	95.17	26.49	91.30	79.60

Table 20: Poem-level base-meter accuracy (%) by base meter. N is the number of held-out test poems associated with each meter. All systems are evaluated on the same aligned test instances, and every instance is included. The highest accuracy in each row is bolded.

Length Bucket	N	Base-Meter Accuracy (%)				Count Accuracy (%)	
		SHAER	Yehia-7B	Ashaar	Fanar-Diwan	SHAER	Yehia-7B
1–3 verses	1,668	97.00	27.04	91.13	78.60	98.44	72.54
4–6 verses	805	93.66	23.98	91.18	79.13	93.17	36.02
7–10 verses	473	93.02	29.60	92.39	80.76	75.26	9.51
11–20 verses	535	93.64	25.79	91.03	82.43	28.97	0.56
Overall	3,481	95.17	26.49	91.30	79.60	83.40	44.47

Table 21: Poem-level base-meter and exact count accuracy by requested poem-length bucket. Count accuracy is reported only for SHAER and Yehia-7B because Ashaar and Fanar-Diwan do not support requested length conditioning. Every applicable held-out instance is included.